

Time: 25 Minutes

You have implemented a Smart Irrigation System using Laplace Transform. Now you will extend your implementation to model real-world pump degradation and sensor noise, and analyse additional performance characteristics.

Task 1: Error Function

Add a method `u_error_A(u)` that takes any input signal and returns a corrupted version:

$$u_corrupted(t) = u(t) + e(t)$$

where the error is a slow pump degradation combined with periodic vibration:

$$e(t) = -0.02 \cdot t + 0.08 \cdot \sin(8t)$$

Here,

$-0.02 \cdot t$ models gradual wear reducing pump efficiency over time

$0.08 \cdot \sin(8t)$ models high-frequency mechanical vibration in the pump motor

Apply this to the Step input and Exponential input only.

Task 2: New Metrics

Implement the following two methods:

A. *Integral of Squared Error (ISE)*

$$ISE = \int (h_{ss} - h(t))^2 dt$$

where h_{ss} is the steady-state value of the output signal.

B. *Response Energy(E)*

$$E = \int h(t)^2 dt$$

Task 3: Plot

For each of the **two inputs (Step, Exponential)**, produce a figure with two subplots side by side:

Left: Laplace simulation with clean input (original, no error)

Right: Laplace simulation with corrupted input (with error added)

Both plots should show the input signal and output $h(t)$. Print the two new metrics for both the clean and corrupted responses.